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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

CERTIFICATE

This is to certify that Project entitled **"PrintChakra – AI-Powered Smart Print and Capture Solution"** carried out by **CHAMAN S (1VK22AI007), GOWTHAM S (1VK22AI013),** and **KARTHIK KUMAR V GUPTA (1VK22AI016)** bonafide students of **Vivekananda Institute of Technology,** have satisfactorily completed the **Project Phase I (BAI685)** prescribed for 6th semester in **Artificial Intelligence and Machine Learning, Visvesvaraya Technological University, Belagavi** during the year 2024-2025. The Project report has been approved as it satisfies the academic requirements prescribed for the said degree.

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DECLARATION

We hereby declare that Project entitled **"PrintChakra – AI-Powered Smart Print & Capture Solution"** submitted by us, for the award of degree of Bachelor of Engineering in Artificial Intelligence and Machine Learning to Visvesvaraya Technological University is a record of bonafide work carried out by us under the guidance of **Dr. Shaila K**, Professor & HOD, department of Artificial Intelligence andMachine Learning, Vivekananda Institute of Technology. We further declare that the work reported in this Project has not been submitted and will not be submitted, either inpart or full, for the award of any degree or diploma in this institute or any other university.

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ABSTRACT

In today's rapidly digitizing world, manual document processing remains a bottleneck in educational institutions, offices, and government sectors. Traditional workflows involving physical printing, manual scanning, and text extraction are inefficient, time-consuming, and prone to human error. To address these challenges, this project introduces **PrintChakra – an AI-Powered Smart Print & Capture Solution** that combines AI, mobile computing, and offline synchronization to automate the print-to-digital pipeline.

The system integrates an Android application and a Windows desktop interface to streamline real-time document capture, enhancement, and OCR-based text extraction. Using deep learning-enhanced edge detection via OpenCV, the mobile app automatically captures and corrects document images, while Tesseract OCR extracts text data with high accuracy. A Flask-based REST API enables seamless offline synchronization between the mobile and desktop applications over a local Wi-Fi network, removing the need for internet access.

Data, including extracted text and document images, is securely stored using MongoDB and GridFS, while JWT-based authentication and SSL encryption ensure secure communication. A React.js-powered dashboard on the desktop allows users to view scanned documents, access extracted content, and manage files in real time.

The project involved challenges like accurate edge detection, OCR tuning for different fonts and lighting conditions, and real-time offline communication, all addressed through AI model tuning, multi-threaded processing, and adaptive scanning techniques.

Achieving reliable synchronization and OCR performance, the system significantly reduces manual effort, enhances document processing speed, and enables intelligent record-keeping. Future improvements may include handwriting recognition, language auto-detection, and cloud integration for remote access and scalability.

PrintChakra marks a leap forward in document automation, replacing fragmented manual workflows with a secure, intelligent, and autonomous print-and-capture system tailored for modern institutions and enterprises.

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CHAPTER 1

INTRODUCTION

Document management plays a critical role in the daily operations of educational institutions, corporate offices, and government organizations. Traditional methods involving manual printing, scanning, and text entry are not only time-consuming but also susceptible to errors and inefficiencies. These workflows often require multiple tools and human intervention at every stage — from placing a print command to scanning physical documents, enhancing the image, extracting content, and finally storing the results. The lack of automation significantly hampers productivity, especially in environments that demand large-scale documentation and digitization.

To address these inefficiencies, **PrintChakra – an AI-Powered Smart Print & Capture Solution** is designed to automate the end-to-end process of physical document handling. This intelligent system integrates AI-based image enhancement, Optical Character Recognition (OCR), and offline synchronization to bridge the gap between physical and digital documents. Leveraging OpenCV for real-time edge detection and perspective correction, combined with Tesseract OCR for accurate text extraction, the system allows users to capture printed content directly through an Android smartphone and transfer it seamlessly to a Windows desktop application over a local network.
Unlike traditional workflows that depend on internet connectivity or multiple standalone applications, PrintChakra functions entirely offline using a secure Wi-Fi network. MongoDB and GridFS handle the backend storage of large image and text files, while JWT and SSL ensure secure communication. A React.js-based dashboard allows real-time document monitoring, editing, and management from the desktop interface.

By embedding AI directly into the capture and processing stages, this project not only reduces manual workload but also enhances speed, accuracy, and data integrity. PrintChakra transforms conventional document workflows into a streamlined, intelligent, and autonomous system, paving the way for efficient document automation across diverse sectors.

1.1 CHALLENGES

1 **Hardware Interfacing**: Establishing offline connectivity between mobile, printer, and desktop requires precise setup and synchronization.

2 **Real-Time Processing**: Mobile devices have limited resources for edge detection, correction, and OCR, needing optimized algorithms.

3 **OCR Reliability**: Varying fonts, formats, lighting, and paper quality affect OCR accuracy and output clarity.

4 **Offline Sync**: Ensuring fast, secure data transfer over local networks introduces latency and sync issues.

5 **Storage Scalability**: Managing large image/text data in GridFS while ensuring quick access demands efficient design.

6 **Security**: Sensitive document handling requires robust SSL and JWT-based API protection to ensure data privacy.

1.2 MOTIVATION

In a world increasingly driven by information, efficient document management is critical. Yet, in many institutions and workplaces, traditional print-scan workflows remain manual, time-consuming, and prone to errors—leading to productivity loss and data inconsistencies.

Employees often spend valuable time printing, scanning, and manually entering information, especially in environments with limited internet connectivity or legacy systems. These inefficiencies highlight the need for a smarter, more integrated approach.

PrintChakra is motivated by the vision to bridge the gap between physical documents and digital systems. By leveraging advancements in AI, image processing, and OCR, the project aims to automate document capture, extraction, and syncing—all in real-time and without internet dependency.

This solution empowers users with a seamless, secure, and intelligent document workflow—enhancing accuracy, reducing human effort, and adapting to real-world constraints like offline environments and hybrid device usage. It's about transforming everyday document tasks into a streamlined, smart experience.

1.3 PROBLEM STATEMENT

Traditional document workflows involving physical printing, manual scanning, and data entry are inefficient, labor-intensive, and error-prone. These systems lack automation, require continuous human involvement, and are not scalable for institutions handling large volumes of documents. Moreover, the dependency on internet connectivity and multiple disconnected tools adds friction to the process. This project aims to develop an AI-powered smart print-and-capture solution—**PrintChakra**—that automates real-time document capture, image enhancement, and OCR-based text extraction. By integrating mobile-based scanning, AI-driven image processing, and offline synchronization with a desktop system, the solution will improve productivity, accuracy, and operational efficiency in document handling.

1.4 OBJECTIVES

- To automate the capture and digitization of printed documents using an Android-based scanning application.
- To apply AI-based edge detection and perspective correction for enhanced image clarity and accuracy.
- To implement OCR using Tesseract for reliable text extraction from captured images.
- To enable secure offline synchronization between the mobile app and Windows desktop via Flask APIs.
- To design a React.js dashboard for viewing scanned documents, managing extracted text, and generating printable reports.
- To ensure secure data handling through JWT-based authentication and SSL-encrypted communication.
- To build a scalable backend using MongoDB and GridFS for efficient document storage and retrieval.

1.5 SCOPE OF THE PROJECT

- **Automated Document Capture**: Enable on-demand and triggered scanning of printed documents using AI-enhanced mobile capture, eliminating the need for manual scanning
- **AI-Enhanced Processing:** Use OpenCV for edge detection and Tesseract OCR for high-accuracy text extraction under various lighting conditions and document types.
- **Real-Time Processing and Syncing**: Perform edge detection, perspective correction, and OCR in real-time, with immediate syncing between mobile and desktop.
- **Intelligent Storage Management**: Automatically store extracted text and images in MongoDB/GridFS while managing space by avoiding redundant captures.
- **User-Friendly Dashboard**: Develop an intuitive React.js-based interface for monitoring scans, editing text, organizing files, and exporting documents.
- **traction under various lighting and document types.**
- **Multi-Device Synchronization:** Ensure seamless coordination between Android mobile and Windows desktop apps over a local Wi-Fi network for offline workflows.
- **Edge Device Optimization:** Perform all scanning and processing directly on the mobile device to enable offline functionality and reduce latency.
- **Automated Document Logging:** Log and timestamp every scan, extracted text, and user action for audit and tracking purposes.

1.6 ORGANISATION OF THE REPORT

Chapter 1 introduces the project by outlining the challenges of traditional document workflows and the need for automation in print-and-scan environments. It highlights the inefficiencies of manual processes such as scanning, text entry, and synchronization, and sets the foundation for PrintChakra as a smart, AI-powered solution. The chapter also covers the motivation behind the project, clearly defines the problem statement, and lists specific objectives aimed at building an offline, intelligent, and secure document capture system.

Chapter 2 presents a thorough literature survey, exploring existing technologies in image processing, OCR, offline synchronization, and mobile-desktop integration. It reviews research studies related to edge detection algorithms, multilingual OCR, and AI-based enhancements in document capture. The survey also includes techniques like OpenCV for perspective correction, Tesseract for text recognition, and Flask APIs for offline communication, identifying the gaps in current systems that PrintChakra aims to address.

Chapter 3 focuses on the system requirements, detailing both functional and non-functional aspects of the proposed solution. It specifies hardware components such as Android smartphones, LaserJet printers, and desktop systems, along with software stacks including React Native, Python, MongoDB, and GridFS. The chapter also outlines performance goals, scalability expectations, data privacy needs, and system constraints to ensure the final implementation is robust and practical for real-world use.

Chapter 4 describes the architectural design of the PrintChakra system. It includes detailed block diagrams, workflow explanations, and component interactions, illustrating how the mobile app, desktop application, database, and APIs work together. Special focus is given to modularity, real-time synchronization, and offline capability, ensuring that the system remains efficient and extensible for future upgrades such as handwriting recognition and cloud integration.

Chapter 5 explains the development and implementation methodology followed throughout the project lifecycle. It outlines the step-by-step workflow—from hardware setup and mobile app development to API creation, OCR testing, and web dashboard integration. This chapter also discusses the challenges encountered, including lighting variability, OCR accuracy, and offline syncing, and explains how they were resolved through AI optimization, multi-threading, and adaptive processing techniques.

CHAPTER 2

LITERATURE SURVEY

Liu et al. [1] proposed a low-profile inkjet-printed interconnect solution for high-speed and compact hardware interfacing, addressing the limitations of traditional large-form communication architectures. Their work has significant implications for embedded systems where compactness, precision, and latency-free communication are essential. In PrintChakra, this ensures that print jobs and scan responses can be initiated and acknowledged without delay, mimicking real-time hardware interactions in standalone environments such as examination halls and print labs.

Hämäläinen et al. [2] explored the power of deep neural networks (DNNs) in OCR systems, particularly emphasizing the importance of multilingual support, adaptive training, and post-processing optimization. This aligns closely with PrintChakra's integration of the Tesseract OCR engine, which must deal with a variety of document types often printed in different Indian languages, fonts, and formats. By building on their findings, PrintChakra enhances institutional accessibility, enabling scanning of vernacular medium documents and region-specific formats that traditional OCR engines often struggle to decode.

Yasin et al. [3] investigated decentralized P2P systems for secure and reliable file sharing. Although PrintChakra doesn’t adopt a full-fledged P2P framework, the principles of decentralized, peer-to-peer architecture resonate in its offline-first, localized design. This ensures users are not dependent on central cloud services or constant internet availability, which is particularly vital in areas with restricted or sensitive data handling policies.

Frei et al. [4] contributed to foundational work on real-time edge detection using orthogonal polynomial-based methods. Their influence is evident in PrintChakra’s OpenCV-based preprocessing, which includes document localization and noise filtering. This reduces the need for manual cropping or image correction, making mobile capture significantly faster and user-friendly, especially when scanning multiple documents in succession.

Li et al. [5] expanded on conventional edge detection by creating adaptive models capable of operating effectively in noisy or variable environments. This is directly applicable to PrintChakra, which operates in real-world conditions such as indoor classrooms or offices with mixed lighting. The adaptive nature allows the app to self-tune contrast and sharpness dynamically, improving OCR outcomes even when image quality is suboptimal.

Mansouri et al. [6] evaluated the performance of distributed NoSQL databases in edge computing environments, with a focus on bandwidth, latency, and energy efficiency. Their conclusion supports PrintChakra's use of MongoDB with GridFS for storing scanned images and extracted text efficiently. The ability to query document metadata without loading large image blobs boosts performance and makes the dashboard experience fluid, even when thousands of documents are stored.

Siripurapu et al. [7] compiled a taxonomy of synchronization protocols, discussing their suitability under various offline and edge use cases. Inspired by this, PrintChakra employs an acknowledgment-based REST sync system with error handling and retry mechanisms. This not only maintains file integrity across devices but also prevents data duplication or corruption, a crucial feature in academic documentation workflows.

Boškov et al. [8] examined trade-offs in sync protocol design, particularly focusing on REST-over-LAN scenarios. Their analysis influenced PrintChakra's approach to minimize overhead by batching requests and reducing unnecessary polling. This results in better battery life on mobile devices and reduced CPU load on desktops, which is important for long-duration scanning tasks in low-resource settings.

Phung et al. [9] addressed REST-based mechanisms for resource-constrained IoT systems with a focus on resiliency and low resource consumption. Their recommendations led PrintChakra to implement stateless and secure Flask-based APIs for mobile-desktop communication. Even when used on older Android phones or in congested Wi-Fi networks, this approach guarantees responsive performance and secure synchronization.

Agrawal et al.[10] enhanced Tesseract's architecture to minimize OCR errors in poor-quality scans. PrintChakra extends this with preprocessing filters and multilingual packs, improving accuracy for complex tables, numbers, and handwritten exam or admission forms.

BACKGROUND

The PrintChakra project arises from the limitations of traditional print–scan workflows, which are often slow, manually intensive, and reliant on internet connectivity. These challenges are magnified in institutional settings such as schools, libraries, and government offices where large volumes of paperwork require reliable processing.

To overcome these constraints, PrintChakra integrates edge AI, offline communication protocols, and robust storage mechanisms. The foundational hardware interfacing strategy mirrors the inkjet-interconnect model proposed by Liu et al. [1], ensuring seamless visual communication between a PC and a printer without cloud mediation.

At the front-end of the workflow, image capture begins on an Android device, where document edges are detected using OpenCV implementations rooted in the theories of Frei and Chen [4], further optimized with adaptive models introduced by Li et al. [5]. These models allow the app to crop and straighten printed material, even under suboptimal lighting, ensuring clean input to the OCR system.

Text recognition is performed using Tesseract, enhanced by insights from Agrawal [10] and Hämäläinen & Haapalainen [2], who proposed DNN-based post-processing for improved multilingual recognition. Extracted data is pushed via secure REST APIs modeled after Boškov et al. [8] & Phung et al. [9] to the Windows backend, where it is logged and indexed.

Crucially, PrintChakra operates offline, with synchronization handled over a Wi-Fi LAN between mobile and desktop systems, drawing on architectural models proposed by Yasin [3] and Siripurapu [7]. This ensures continuous functionality in low-connectivity environments.

Finally, document metadata and OCR outputs are stored using MongoDB's GridFS, validated by the distributed database evaluations from Mansouri et al. [6], providing scalable and efficient storage for large-format scanned documents.

Together, these components make PrintChakra a resilient, intelligent, and offline-capable document automation solution.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 HARDWARE REQUIREMENTS

1. MOBILE DEVICE

The mobile device (Preferably android phone), is the primary input interface of the system, used for real-time document capture through its rear camera. A phone with a high-resolution camera (min 12 MP), good autofocus, and stable performance under various lighting conditions ensures crisp image acquisition for accurate OCR processing. Devices supporting AI hardware acceleration or embedded vision modules are ideal for enhancing edge detection & image preprocessing on the fly. A sturdy tripod or overhead mount is recommended to maintain alignment between the camera & printed documents during scanning.

2. DESKTOP SYSTEM

The desktop or laptop system serves as the central hub for image processing, synchronization, document storage, and user access. It should have a modern multi-core processor (Intel i5 or above), a minimum of 8 GB RAM, and at least 256 GB of SSD storage for smooth performance. It runs the backend application built with Flask, handles incoming data from the mobile device, and initiates operations like OCR conversion, PDF export, and document archiving. Optional GPU support improves processing speeds for advanced OpenCV functions and large batch processing.

3. PRINTER STATION

The printer station in PrintChakra functions as a passive but essential element. It provides a uniform, well-lit surface where printed documents are placed for capture. Any standard LaserJet or inkjet printer with a flat output tray can be used. While not directly connected for scanning or printing purposes, the alignment of documents on the tray ensures consistent framing and stable edges, essential for edge detection and high OCR accuracy during image capture.

4. ROUTER OR HOTSPOT DEVICE

A local Wi-Fi router or mobile hotspot device provides the communication channel between the smartphone and the PC. It supports REST API–based offline data exchange over a local network without the need for internet access. For optimal performance, the router should support at least 802.11n or higher standards, providing stable bandwidth for transferring image and metadata files in real time. LAN support is a bonus for connecting the desktop for faster sync operations and improved reliability.

5. STORAGE HARDWARE

Reliable storage is essential for managing high-resolution document images and their corresponding OCR outputs. A solid-state drive (SSD) is recommended over HDD for faster read/write cycles, especially during real-time image processing and file exports. In long-term or high-volume deployments, external drives or Network Attached Storage (NAS) devices can be integrated for archival and backup. Storage must support organized data structuring, secure access, and backup scheduling for institutional usage.

3.2 SOFTWARE REQUIREMENTS

1. DOCUMENT CAPTURE APP

The Android application built using React Native and OpenCV libraries is responsible for document scanning, edge detection, enhancement, and transmission. It uses AI-powered visual cues to identify document boundaries, auto-capture well-aligned scans, and compresses image files for faster transmission. The app communicates with the desktop using secure REST APIs and supports offline queuing in case of connectivity loss. It also allows users to review and re-scan if needed.

2. BACKEND SERVER SOFTWARE

The desktop backend application is developed using Python with Flask. It exposes REST endpoints to receive scanned images from the mobile app, processes them using OpenCV and Tesseract OCR, and handles conversion to searchable PDFs or text files. The server also logs metadata, manages sync confirmations, and stores processed data in MongoDB/GridFS. It supports multithreading and robust error handling to ensure smooth document workflows.

3. **USER DASHBOARD INTERFACE**

The desktop or browser-based user dashboard, built using React.js, provides access to scanned documents, logs, and system configurations. It enables administrators to view history, manage storage, search records, and export documents. Real-time status updates and sync indicators help monitor ongoing tasks. The interface is responsive and supports role-based access using JWT tokens to protect sensitive data.

4. OCR AND IMAGE PROCESSING TOOLS

The system employs Tesseract OCR for multilingual text recognition and OpenCV for real-time image preprocessing. Functions include brightness normalization, noise filtering, edge detection, and perspective correction. FFmpeg may be used for handling any media format conversion. These tools work in tandem to deliver high-quality text extraction, even from skewed, faded, or shadowed documents.

3.3 COMPUTATIONAL REQUIREMENTS

1. PROCESSING POWER

To support real-time scanning and processing, the desktop system must feature a multi-core CPU (Intel i5/i7 or equivalent Ryzen processor) with optional GPU support for accelerated OpenCV operations. Tasks include receiving images, performing edge detection, running OCR, and exporting results—all with minimal delay. For mobile devices, at least an octa-core processor and 4 GB RAM is required to run the scanning app and perform pre-upload enhancement smoothly.

2. STORAGE CAPACITY

Each document scanned at high resolution requires between 1–5 MB of storage. With hundreds of scans daily in institutional settings, the system must have at least 256 GB SSD for desktop storage. GridFS is used to manage chunked storage in MongoDB, allowing better performance for large image files. External backup or NAS devices may be used for data redundancy and archival.

3. Network Bandwidth

The system uses local Wi-Fi for syncing. Though no internet is needed, local file transfer over Wi-Fi must support at least 5 MB/s to ensure smooth image and metadata exchange without lag. A dual-band router (2.4 GHz and 5 GHz) is preferred for better stability and performance.

3.4 DATA REQUIREMENTS

1.High-Resolution Image Data

Captured images must be of high clarity to ensure accurate OCR results. The mobile app captures images at resolutions between 1080p and 4K, depending on the camera. OpenCV processes these to enhance text legibility before OCR extraction. Noise reduction and contrast enhancement are applied automatically.

2. Text Data (Ground Truth)

Labeled datasets for different fonts, layouts, and languages are necessary to test OCR accuracy. Sample documents with known content are used to evaluate extraction reliability under different lighting and alignment conditions, serving as ground truth during testing.

3. Metadata and Document Logs

Each scan is associated with metadata including timestamp, document type, user ID, and device ID. This is stored alongside text output in MongoDB, enabling advanced search, analytics, and audit trails. Logs also track file transfers and system performance for maintenance and debugging.

4. Security Tokens and Credentials

JWT tokens are required to authenticate users and APIs. All communications between mobile and desktop use encrypted channels (SSL), ensuring the confidentiality and integrity of document data, especially in institutions or corporate environments.

CHAPTER 4

SYSTEM DESIGN AND ARCHITECTURE

4.1 SYSTEM OVERVIEW

PrintChakra is an AI-powered solution built to automate and simplify document digitization in environments where traditional scanning methods are inefficient or disconnected. It enables real-time capture, processing, and synchronization of printed materials into searchable digital formats—without relying on internet connectivity.

The system consists of two main components: an Android-based mobile app and a Windows-based desktop application, connected via secure RESTful APIs over local Wi-Fi. The mobile app uses the phone camera to scan documents, leveraging OpenCV for edge detection, perspective correction, and image enhancement in real time. Once processed, the images are compressed and securely transmitted to the desktop module.

On the desktop, additional enhancements are applied, followed by text extraction using Tesseract OCR. The extracted content and metadata such as timestamps and device/user IDs are stored efficiently in MongoDB using GridFS, which handles both large images and structured data.

A user-friendly, role-based React.js dashboard allows users to manage scanned documents—reviewing, editing, organizing, and exporting them as searchable PDFs. JWT-based authentication ensures secure access, making it suitable for sensitive or institutional use cases. Optimized for offline-first and low-bandwidth environments, PrintChakra supports multiple mobile devices connecting to a single desktop hub, with future-ready provisions for cloud integration and multi-user scalability.

By combining mobile AI processing, secure offline syncing, and modular architecture, PrintChakra eliminates the need for traditional scanners and cloud-reliant apps—offering a smart, secure, and autonomous document management solution.

4.2 ARCHITECTURAL COMPONENTS

1. DOCUMENT CAPTURE MODULE

The Document Capture Module is implemented as a mobile application that uses the Android device's rear camera to scan physical documents. It incorporates OpenCV-based edge detection, contrast enhancement, and auto-cropping features to capture clean, high-resolution document images. The app assists users through visual guides and auto-snap functionality, ensuring that the document is aligned properly during capture. It also includes image preprocessing techniques like brightness normalization and noise filtering, improving OCR accuracy. Once captured, the image is temporarily stored and prepared for transmission to the desktop backend. The capture module is optimized for speed and minimal interaction, reducing scanning time and manual correction.

2. IMAGE TRANSMISSION AND SYNC MODULE

This module establishes a secure, offline communication channel between the mobile device and the desktop system using Flask-based REST APIs over a local Wi-Fi network. Images captured by the mobile app are encoded and transmitted as HTTP POST requests to the server. The sync module features retry logic, data validation, and response verification to ensure reliable delivery even under unstable connections. It queues unsent files locally when offline and transmits them automatically when the connection is restored. The communication layer supports lightweight encryption and uses JWT authentication to validate each request, ensuring that only authorized devices can interact with the backend.

3. OCR AND TEXT PROCESSING ENGINE

Once the image is received by the desktop system, it is passed to the OCR and Text Processing Engine. This module utilizes OpenCV for further image enhancement, including resizing, sharpening, and binarization. Tesseract OCR is used for extracting text from the processed image, with support for multiple languages and mixed fonts. The engine converts the raw scan into searchable text and optionally generates PDF or .txt output formats. Custom pre- and post-processing logic is implemented to handle layout retention, error correction, and noise filtering, improving the usability and readability of the extracted content.

4. MONGODB STORAGE MODULE

The Storage Module employs MongoDB with GridFS to store large document images alongside their OCR-extracted data. GridFS divides files into manageable chunks, enabling efficient storage and retrieval of large files. Each entry in the database includes the original image, the text output, metadata (timestamp, device ID, document type), and user-defined tags. The structure supports indexing and fast querying, enabling administrators to search for documents by keyword, date, or category. MongoDB's NoSQL architecture ensures flexibility and scalability, making it ideal for institutions or multi-user environments with growing document repositories.

5. DOCUMENT MANAGEMENT AND ALERT SYSTEM

This module is responsible for tracking scan activity and optionally alerting users when uploads or OCR tasks fail or succeed. It maintains logs for each transaction, noting timestamps, processing status, and any errors encountered. Notifications are pushed to the user dashboard in real time. While SMS or email alerts are optional, the system supports integration with external services (like SMTP or Firebase Cloud Messaging) for customizable notifications. This ensures visibility into system behavior and provides accountability for document processing in sensitive environments like schools or offices.

6. **WEB INTERFACE AND ADMIN DASHBOARD**

The user interface is built as a web-based dashboard using React.js for the frontend and Flask for the backend API. It offers a centralized workspace where users can view scanned documents, manage metadata, download OCR outputs, and search records using filters. The dashboard supports login authentication, user management, and role-based access control. Admins can monitor processing logs, update system configurations, and view sync status between mobile and desktop. Designed with usability in mind, the dashboard includes mobile-responsive views, real-time progress indicators, and PDF preview functionality for document verification and export.

4.3 SYSTEM ARCHITECTURE

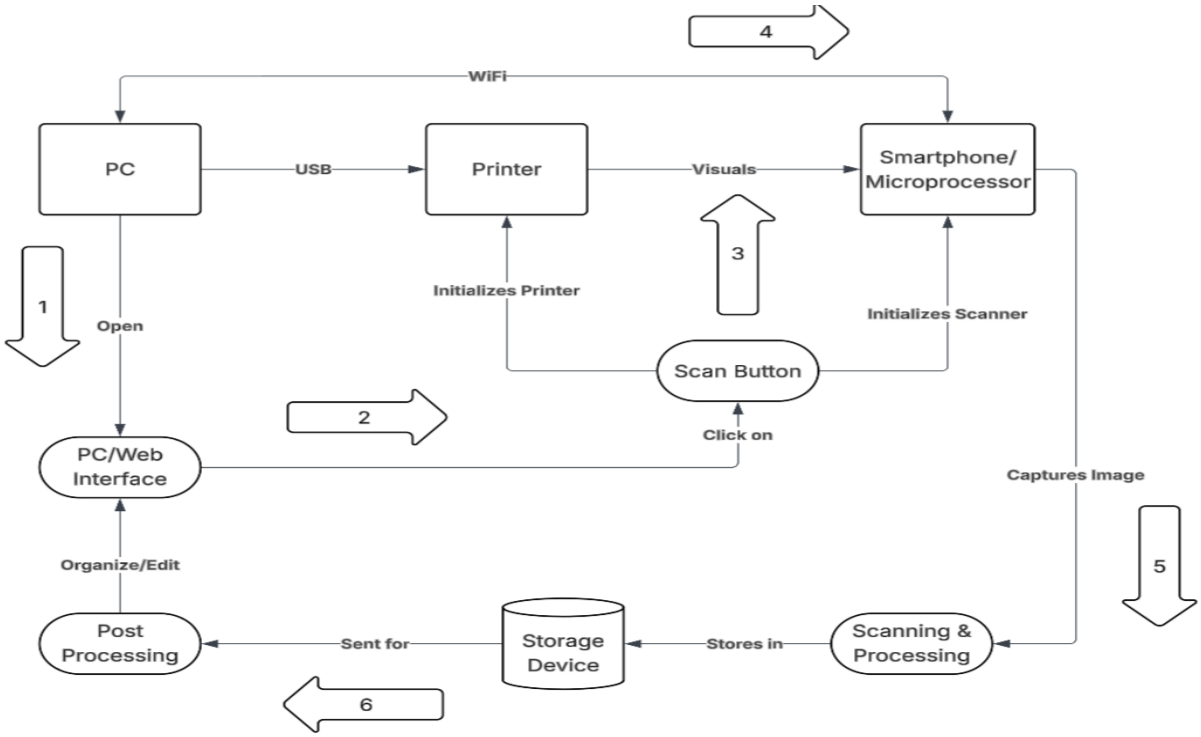

Figure 4.1: Block Diagram

The above diagram outlines the high-level workflow of the **PrintChakra – AI-Powered Smart Print & Capture Solution**, designed for intelligent, offline-first document scanning and synchronization between a mobile device and a PC.

The process begins when the user powers on the **PC** and then opens the **PC/Web Interface** (Step 1). Through this interface, users can access previous documents, adjust OCR settings, or prepare the system for new scans.

Once the interface is active, the user initializes the printer (via **USB** or **Wi-Fi**) to prepare a document for capture (Step 2). The **Printer** visually outputs the document, which is then captured by a **Smartphone or Microprocessor Device** (Step 3), typically running the PrintChakra mobile scanning application. The user presses the **Scan Button** on the app, which activates the camera and initializes the AI-driven scanner module.

After the **document is visually captured**, the system enters the **Scanning & Processing** stage (Step 4), where the image undergoes preprocessing (e.g., edge detection, de-skewing) followed by **OCR** to extract text data. This processed output is then securely stored in the **Storage Device** on the PC (Step 5), usually managed via MongoDB and GridFS.

Finally, the stored document is available for **Post Processing** and organization (e.g., tagging, exporting, reviewing), which can be done through the **PC/Web Interface**. Users can search, filter, or export results as needed (Step 6), completing a full offline-to-digital processing cycle.

This block diagram demonstrates how PrintChakra enables seamless coordination between hardware (printer, smartphone, PC) and software components (OCR, REST APIs, database), without needing cloud connectivity. Its design supports offline scanning, fast syncing, and secure document management for educational, institutional, or remote work environments.

4.4 SYSTEM FLOW

1. Launch and Web Interface Initialization

- The user opens the PrintChakra web interface on the PC to access or manage previous scans.
- The interface initializes internal modules such as OCR configuration, metadata templates, and database connections.
- Admin authentication (if enabled) ensures secure access to stored documents and settings.

2. Document Preparation and Printer Initialization

- The connected printer (via USB or Wi-Fi) is initialized and used to produce a physical copy of the document intended for capture.
- The user positions the paper for optimal visibility from the smartphone camera.
- Alternatively, a pre-existing printed document may be used.

3. Image Capture via Mobile Application

- The user launches the PrintChakra mobile app and taps the Scan Button to activate the device camera.
- The application uses OpenCV-based edge detection and perspective correction to align and frame the document.
- Real-time feedback is provided for clarity, alignment, and lighting conditions.
- Once the document is captured, it is passed to the image processing module on the mobile or PC side, depending on the sync configuration.

4. Image Preprocessing and OCR

- Captured images are sent to the desktop server over local Wi-Fi using Flask-based REST APIs.
- The server receives the file, applies additional image filters, and passes it through Tesseract OCR to extract structured text.
- The OCR module supports layout-aware extraction, helping retain formats like bullet points, titles, and tables.

5. Storage and Metadata Logging

- The output text, along with the original image and metadata (timestamp, user ID, file type), is saved in a MongoDB GridFS storage system.
- A backup entry is created for reference, allowing users to later search or filter based on keywords, date, or tags.

6. Post Processing and Export

- Documents are made available for review, editing, tagging, or exporting via the PC/Web interface.
- Users can organize scanned files, apply filters, or export OCR results in PDF, TXT, or Excel format.
- Version control ensures that any modifications are logged and retrievable.

7. User Features and Dashboard Controls

- The web dashboard includes a scan history log, showing recent activities, timestamps, and OCR success status.
- Users can apply filters such as file type, scan date, or custom tags for quick retrieval.
- Admins can manage user roles, export activity reports, and configure system-level settings (like sync paths, resolution preferences, or OCR languages).
- The system is designed for offline use, but includes optional cloud sync modules for future upgrades or hybrid deployments.

4.5 COMPONENTS AND TECHNOLOGIES USED

COMPONENT	TECHNOLOGY USED
Image Capture	OpenCV, React Native Camera, Android Camera2 API
Edge Detection & Processing	OpenCV, NumPy
OCR Engine	Tesseract OCR, Pillow (PIL)
Backend Server	Flask (Python)
Frontend Dashboard	React.js, HTML, CSS, Bootstrap
Database	MongoDB, GridFS
Communication (Sync)	REST API (Flask), JWT, HTTPS
Authentication & Security	JWT Tokens, SSL
System Deployment	Python, MongoDB, Ubuntu, Local Wi-Fi or Hotspot

Figure 4.2: Table of Components and Technologies used

The table above outlines the core technologies that form the foundation of the PrintChakra system. The **image capture module** uses React Native with Camera2 API and OpenCV to capture and preprocess high-resolution document scans via the Android device. **Edge detection** and perspective correction are performed using OpenCV and NumPy libraries to ensure proper framing before OCR.

The **OCR engine** is powered by Tesseract OCR, integrated with Python's Pillow library for image handling and pre-processing tasks such as grayscale conversion and noise removal. The **backend server** is developed using Flask, a lightweight and flexible Python framework that handles image reception, processing, and data storage workflows.

The **frontend dashboard** is developed using React.js, along with HTML, CSS, & Bootstrap to provide a responsive and intuitive user interface for reviewing scanned documents, managing metadata, and exporting outputs. MongoDB is used for backend data storage, while **GridFS** handles the large binary files, enabling efficient image and document management.

For communication between mobile and desktop, the system uses a secure **REST API**, protected with **JWT authentication** and SSL encryption. This ensures safe, offline-first synchronization over local Wi-Fi or hotspot networks.

Deployment of the system is done using **Python** on **Ubuntu-based desktops or laptops**, with MongoDB running as the local data store. The system is designed for offline operability, and can function independently in secure, air-gapped environments, such as government offices or academic institutions with restricted internet access.

4.6 SECURITY AND PRIVACY CONSIDERATIONS

1. Authentication

PrintChakra secures system access using encrypted user credentials, stored via bcrypt or SHA-256 hashing in MongoDB. Login sessions are managed through JWTs, which include expiry times and signature verification to prevent unauthorized access. This ensures that only authenticated users can interact with protected system components.

2. Encrypted Data Transfer

All data exchanged between the mobile app and desktop backend is secured using HTTPS and SSL encryption. This protects REST API communication during offline Wi-Fi synchronization, ensuring that scanned documents and OCR metadata remain confidential and tamper-proof during transfer.

3. Encrypted Storage

PrintChakra uses encryption at both file and field levels. Document images are stored in MongoDB GridFS as encrypted chunks, while text and metadata are protected using Fernet encryption. Even if local storage is compromised, unauthorized users cannot access meaningful data without the decryption keys.

4. Role-Based Access Control (RBAC)

The system implements RBAC to restrict features based on user roles. Admins have full access to manage users, documents, and system configurations, while regular users are limited to scanning, syncing, and viewing their own documents. This protects critical operations from unauthorized access or misuse.

5. Data Retention and Logging

PrintChakra logs all major activities—logins, file uploads, OCR executions, and exports—with secure admin-only access. The system also supports configurable data retention policies, helping to regularly clean outdated files, reduce storage load, and minimize security exposure.

CHAPTER 5

METHODOLOGY

LAYING THE FOUNDATION FOR EFFICIENT AUTOMATION

Building a robust and intelligent document capture and processing solution like PrintChakra begins with setting up a structured and well-optimized development environment. This foundational phase plays a critical role in ensuring consistent performance, modular design, and seamless integration across devices. Just like a well-laid foundation determines the stability of a building, the design of the development environment directly influences system reliability, deployment scalability, and real-time responsiveness.

A thoughtfully configured setup enables rapid testing, debugging, and adaptation of core components such as edge detection, OCR pipelines, file synchronization, and document management. It also allows for efficient evaluation of AI performance in varied real-world scenarios, such as document misalignment, lighting inconsistencies, and network fluctuations in offline settings.

The environment includes several core components critical to the success of the system:

1. Programming Tools & Languages:

Python is the core development language due to its extensive ecosystem for computer vision, web APIs, and backend development. Flask is used for the backend REST API, while JavaScript (React.js) is used for the user dashboard. React Native is adopted for the cross-platform mobile application. Together, these tools allow developers to build scalable, maintainable modules across both the desktop and mobile environments.

2. Image Processing & OCR Libraries:

OpenCV forms the backbone of image preprocessing functions like document edge detection, noise reduction, and perspective correction. Tesseract OCR is used for accurate text extraction from scanned images, supporting multilingual input and text layout preservation. These libraries allow real-time conversion of printed content into searchable digital formats, even under inconsistent visual conditions.

3. AI-Powered Enhancements:

While the current system leverages rule-based preprocessing and OCR, it is designed for easy integration of deep learning enhancements. This includes optional AI modules for improved edge detection or layout classification using TensorFlow Lite or ONNX models in future iterations.

4. Database and File Management:

MongoDB, combined with GridFS, handles large document images and associated text or metadata. This schema-less database structure allows flexible indexing and quick querying of scanned records, user logs, and synchronization status. File chunks are stored efficiently, ensuring stability during high-volume data operations.

5. Deployment & Platform Integration:

The system is developed and tested on Ubuntu-based desktops and Android mobile devices, using Python-based deployment scripts. Local Wi-Fi is used to simulate offline synchronization, while Flask APIs manage data exchange between the smartphone and the desktop. Docker may be used in future iterations to containerize services for simplified cross-platform deployment and scalability across departments.

By grounding the development process in a modular, AI-ready, and offline-first methodology, PrintChakra is positioned to deliver a reliable, intelligent, and scalable print-to-digital solution that works seamlessly across varied institutional settings.

CORE DEVELOPMENT TOOLS & TECHNOLOGIES

1. Programming Language & IDE

• Python:

Python serves as the backbone of PrintChakra's backend system due to its extensive support for OCR, image processing, and web frameworks. Libraries like OpenCV, Tesseract, and Flask make it a perfect choice for building AI-driven, modular applications. Python's readability and rich package ecosystem enable rapid prototyping and integration with cloud or local data pipelines.

• JavaScript (React & React Native):

JavaScript powers both the frontend admin dashboard and the mobile scanning interface. React is used for building a responsive web dashboard, while React Native provides cross-platform support for Android-based mobile document capture. The component-driven architecture of JavaScript frameworks ensures modular, maintainable UI development.

• HTML, CSS, and Bootstrap:

These technologies are used in combination with React.js to create a modern and intuitive user interface. Bootstrap enhances responsiveness and layout consistency, making the dashboard accessible on multiple screen sizes.

• Popular IDEs:

• VS Code (Python, JavaScript):

Lightweight and highly customizable, Visual Studio Code is used for both backend and frontend development. Its built-in terminal, Git integration, and Python/JavaScript extensions streamline cross-stack development.

• Android Studio (React Native):

Used for developing and testing the mobile capture application. Android Studio provides emulation, device deployment, and debugging tools critical for app reliability.

• MongoDB Compass:

A graphical interface for managing the GridFS document database, Compass allows developers to inspect documents, manage collections, and monitor query performance during testing and deployment.

2. REST API Development & Offline Sync

• Flask RESTful API:

The REST API, built using Flask, is responsible for receiving scanned images, triggering OCR processing, and sending responses to the mobile client. The use of lightweight APIs over local Wi-Fi allows real-time, offline-first communication between the mobile app and the desktop server.

• JWT Authentication:

To ensure secure interactions, the REST APIs are protected using JSON Web Tokens (JWT). Tokens validate user identity, manage permissions, and prevent unauthorized data sync or access.

• Offline Synchronization Logic:

The mobile app is designed to queue unsent scans during disconnections and automatically transmit them when the network is restored. The sync module supports retries, integrity checks, and time-stamped logs for each document transfer.

3. OCR, Image Processing, and AI Integration

• OpenCV:

Used extensively on both mobile and desktop for edge detection, noise filtering, and image transformation. Features like contour detection and adaptive thresholding help ensure accurate document capture even in non-ideal conditions.

• Tesseract OCR:

Integrated on the backend for extracting text from scanned images. PrintChakra uses custom configurations and preprocessing steps to enhance recognition accuracy across various fonts, layouts, and languages.

• Pillow (PIL):

Python Imaging Library (Pillow) is used for image conversion and manipulation tasks like resizing, cropping, and format optimization before OCR is applied**.**

• Optional Deep Learning Add-ons:

Future versions may include lightweight AI models using ONNX or TensorFlow Lite for smarter classification, auto-alignment, and handwriting recognition, especially in academic settings.

4. Version Control with Git

Git is used as the version control system to manage the entire PrintChakra codebase, spanning the mobile app, desktop backend, and frontend dashboard. It ensures every code change, bug fix, or new feature is traceable and reversible. Using Git allows for organized collaboration, streamlined feature branching, & rollback of unstable releases.

• GitHub:

Primary repository host for the project. It includes issue tracking, pull request workflows, and actions for automated testing or linting.

• GitLab/Bitbucket:

Can be integrated for teams already using Atlassian-based environments or for self-hosted CI/CD pipelines.

Git provides the development team with agility and traceability, supporting PrintChakra's iterative improvement and secure release cycles.

SETTING UP THE ENVIRONMENT & UI CONSIDERATIONS

Setting up a well-structured development environment is essential for the reliable and efficient development of PrintChakra. As a cross-platform, AI-powered, and offline-first application, it involves both mobile and desktop components. Proper setup ensures seamless testing, integration, and deployment, while also allowing developers to address platform-specific requirements early in the development cycle.

1. Local Development Environment

The first step involves preparing the desktop development environment with all necessary dependencies and tools. Python 3.x is used as the core programming language for the backend server, image processing, and OCR modules. A virtual environment (using venv or virtualenv) is recommended to manage project-specific packages such as Flask, Tesseract, OpenCV, Pillow, and MongoDB drivers. This prevents conflicts with global installations and allows for cleaner dependency management.

For mobile development, Android Studio is configured with React Native CLI and Node.js. React Native dependencies and plugins for camera access, file transfer, and image preprocessing are installed via npm or yarn within isolated project directories. This dual-environment setup ensures that both mobile and desktop components can be developed and tested independently.

Version control is managed using Git, with the entire project structured into logical modules for UI, backend, database, and sync logic. Integrated development environments (IDEs) such as Visual Studio Code (for Python and JavaScript) and Android Studio (for mobile) support linting, real-time debugging, and terminal operations to boost productivity and reduce development errors.

2. Device Integration (Real or Simulated)

To fully test the functionality of PrintChakra, both real and simulated hardware environments are used. For scanning, a physical Android smartphone with at least 12 MP camera is used for realistic image capture scenarios. The app is mounted above a printer tray or flat surface to replicate actual use conditions involving skewed alignment, lighting variation, and background noise.

On the desktop side, Ubuntu-based laptops or PCs are configured with Flask, MongoDB, and necessary OCR/image libraries. A local Wi-Fi router or hotspot is set up to simulate an offline-only environment. This allows testing of real-time document sync, offline API transfer, and retry logic without relying on internet connectivity.

If mobile devices are not available during early-stage development, Android emulators with camera simulation plugins can be used. However, real device testing is strongly preferred to evaluate camera behavior, resolution handling, and edge detection under practical conditions.

3. User Interface (UI) Considerations

PrintChakra offers two main user interfaces: a mobile app UI for document capture and a desktop web dashboard for reviewing and managing scanned files.

The mobile UI is developed using React Native and emphasizes usability with features like auto-capture, real-time edge detection feedback, and scan history. The interface is kept clean and intuitive, using minimal controls to guide users during scanning without requiring technical expertise.

The desktop dashboard is built using React.js and Bootstrap. It offers responsive layouts for document review, OCR text viewing, and PDF export. Admin controls such as sync status, file search, and metadata filtering are organized in collapsible panels and accessible menus. JWT-based login protects access, and role-based permissions ensure that users only see what they're authorized to view.

Design principles such as consistency, error feedback, and accessibility are followed to ensure a smooth user experience across different devices and screen sizes. The dashboard is optimized for both mouse and keyboard use, supporting institutional staff working with high document volumes.

TESTING & DEPLOYMENT CONSIDERATIONS

Testing Frameworks

Comprehensive testing plays a vital role in ensuring the stability and reliability of the PrintChakra system, especially as it involves multi-device synchronization, real-time processing, and OCR accuracy. The testing process spans across unit testing, integration testing, and simulated end-to-end scenarios to validate system workflows from document capture to storage and export.

For the backend Flask API, OpenCV image handling, and OCR modules, Python-based testing frameworks are utilized:

- **unittest** – Python's built-in testing module is used for validating core functions like edge detection, REST API responses, and file processing routines.
- **pytest** – Offers simplified syntax and powerful fixtures, making it ideal for testing the OCR pipeline, MongoDB interactions, and error-handling logic.
- **coverage.py** – Ensures that all critical paths of the application are tested, identifying untested code segments for improvement.

The React Native mobile app uses tools like Jest for unit tests (e.g., camera permission handling, image validation) and Detox for end-to-end testing of document capture workflows.

Automated testing is executed during every development cycle to catch regressions, validate new features, and ensure stability across offline syncing scenarios and cross-platform interactions.

Logging and Monitoring

Robust logging is integrated into both mobile and desktop components to facilitate issue tracing, performance analysis, and user behavior auditing. The backend server leverages Python's built-in logging module to record events such as scan receipts, sync attempts, OCR errors, and export actions.

Log levels (INFO, WARNING, ERROR) are configured to distinguish routine events from critical issues. Logs can be written to local files and rotated regularly to conserve space, or forwarded to external log processors like Graylog or ELK Stack in enterprise deployments.

For runtime monitoring, lightweight dashboards built using Prometheus + Grafana or custom admin panels can track system uptime, sync latency, image processing times, & error frequencies. These insights help administrators preemptively identify system bottlenecks and maintain operational efficiency.

In the mobile app, critical failures such as image capture failure or upload interruptions are logged locally and optionally synced to the backend once connectivity resumes.

Deployment Considerations

Deploying PrintChakra involves setting up the desktop server and mobile app in real-world environments such as schools, offices, or scanning stations. Key deployment strategies include:

- **Persistent Backend Service:** The Flask server is deployed as a persistent service using systemd or Supervisor on Ubuntu, ensuring automatic restarts in case of system failure or reboot.
- **Scheduled Automation:** Tasks such as file cleanup, metadata backup, and periodic export can be handled using cron jobs or Python's schedule module.
- **Environment Replication:** The production environment mirrors the development setup, including Python version, MongoDB configuration, and all dependencies declared in requirements.txt. Virtual environments ensure isolation and consistency.
- **Docker (Optional):** The backend and OCR components can be packaged in a Docker container for standardized deployment across different machines. This also simplifies updates, scaling, and potential migration to cloud platforms in future iterations.

Before deployment, configuration files are tailored per site, including local network credentials, folder paths, and user roles. Final testing is performed on live devices to ensure seamless sync, stable performance, and full offline functionality.

CHAPTER 6

CONCLUSIONS AND FUTURE SCOPE

6.1 CONCLUSIONS

The integration of AI-powered tools like OpenCV and OCR into a unified mobile-desktop system marks a transformative shift in document digitization. Traditional print-to-digital workflows—reliant on scanners, USB transfers, or manual input—are slow, error-prone, and ill-suited for fast-paced environments like schools, institutions, and remote offices. **PrintChakra** modernizes this process with an offline-first solution that automates document capture, AI-enhanced preprocessing, text extraction, and structured storage—all using a smartphone and a desktop PC, without dependence on cloud infrastructure.

The mobile app captures documents using the phone camera, applies real-time edge detection and enhancement via OpenCV, and sends compressed images over local Wi-Fi using secure RESTful APIs. The desktop component performs OCR using Tesseract and stores both text and metadata in a MongoDB/GridFS database. This seamless, real-time pipeline enables immediate, searchable digitization—ideal for bandwidth-limited or privacy-sensitive environments.

A React-based dashboard provides role-based access for managing scans, reviewing content, generating exports, and monitoring system performance. The architecture is modular and optimized for mid-range devices, featuring JWT authentication, encrypted communication, and lightweight resource usage.

PrintChakra scales easily—supporting multiple mobile devices syncing with a central desktop hub—creating a distributed yet unified document processing system. It accelerates turnaround time, reduces manual errors, and enhances scan quality with real-time feedback, resulting in higher OCR accuracy and consistent file management.

As organizations shift toward paperless operations and digital audits, PrintChakra fills the gap by offering a secure, intelligent, and fully offline solution. It demonstrates how AI can be practically applied to streamline workflows and enhance document integrity in real-world, constrained environments.

6.2 FUTURE SCOPE

1. **Real-Time Handwriting OCR Support:**
   Extend the system's capabilities to recognize & digitize handwritten documents using AI-based handwriting recognition models such as Transformers for handwritten text datasets.
2. **Cloud and Hybrid Synchronization:**
   Introduce optional cloud-based backup and remote document access for multi-site institutions, enabling scalable deployments with encrypted document syncing and role-aware cloud dashboards.
3. **Intelligent Document Categorization:**
   Integrate machine learning classifiers that automatically label and categorize documents based on content, layout structure, or visual features to simplify storage and retrieval.
4. **Contextual Keyword Highlighting:**
   Enhance post-OCR text viewers by allowing dynamic keyword detection and visual emphasis to help users find relevant content faster in large documents.
5. **Multilingual OCR & Layout Retention:**
   Add support for additional scripts such as Devanagari, Tamil, Urdu, or Arabic, & enhance OCR pipelines to retain complex layouts like tables, headers, or multi-column text.
6. **Decentralized Multi-Device Sync:**
   Enable peer-to-peer document sync among mobile devices in group scanning environments using local mesh networking or Bluetooth to eliminate desktop dependency when needed.
7. **Voice-Assisted Capture and Commands:**
   Incorporate voice instructions or confirmations for hands-free scanning and interaction, improving accessibility and ease of use in scanning booths or academic settings.
8. **AI-Powered Feedback Loop:**
   Develop adaptive models that analyze scan quality in realtime and provide users with feedback on alignment, brightness, blurriness, helping improve accuracy before submission.
9. **Embedded and Portable Systems:**
   Deploy a lightweight version of PrintChakra on Raspberry Pi or Jetson Nano for use in field offices or rural institutions, enabling low-cost, high-efficiency document digitization in mobile units.

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